Forget-Set Concentration as an Optimization-Reliability Risk for NegGrad+: Evidence from BatchNorm Networks

Khalil Bibi CentraleSupélec

Working paper, October 2026 — not peer reviewed

Abstract

The class composition of a deletion request is a reliability risk for unlearning in deployed systems, yet it is rarely controlled. We compare fixed-size forget sets concentrated in one class, whose other examples stay in the retained data, with sets spread across classes. In a BatchNorm CNN on Fashion-MNIST, NegGrad+ was unstable for most concentrated sets and never for stratified ones, and the association held in an exploratory replication that included CIFAR-10. In a prospective study of 68 compositions, the early forget loss depended on concentration beyond what class composition predicts (-0.246 decade per bit of label entropy), and concentration shifted pre-clipping gradient norms far more under NegGrad+ than under a matched random-label objective. Two pre-specified follow-ups ask whether the penalty can be avoided: a bundled BatchNorm intervention (RD64) reduced the retained-utility penalty by 13.7 points, and per-condition tuning left a 1.07-point penalty, but no stable configuration reached the retrained model's forgetting force for 7 of 10 concentrated classes. These are optimization findings for the tested configurations, not evidence of erasure, privacy or a mechanism.

1 Introduction

Approximate unlearning is usually judged by its distance to a model retrained without the forget set D_f. Which examples are forgotten changes how hard unlearning is [1, 2], BatchNorm statistics can distort the evaluation of unlearned models [3], and gradient-ascent methods are known to be fragile [4, 5]. In a deployed system, the forget set is not chosen by the engineer: a deletion request may come from one user, one product category or one label. We therefore ask a narrow reliability question. *At a fixed forget-set size, does concentrating D_f in a single class, while that class remains in the retained data D_r, change the behavior of NegGrad+?*

Our contributions are:

- **A reliability failure and its scope.** Concentrated forget sets were associated with NegGrad+ divergence and collapse in the configurations tested, and the early forget loss depended on concentration beyond a prespecified family of class-combination rules (Section 3).
- **An objective-specific signature.** At the main weight, concentration shifted pre-clipping gradient norms far more under NegGrad+ than under a matched random-label objective; a coarse clipping metric hid the smaller shift.
- **Three attempts to avoid the penalty**: BatchNorm variants, a bundled BatchNorm intervention and per-condition tuning of the learning rate and forget weight, the last two under pre-specified protocols (Section 4), with lessons for deployment (Section 5).

The work began with an unexpected failure and continued with post hoc diagnostics, then with studies whose protocols and analyses were committed before their main runs. Each result carries its status (Table 1). We separate optimization divergence, classification collapse, retained utility and a forgetting-force proxy; none of them measures erasure or privacy, and we claim neither, nor a mechanism or generality beyond the models and data tested.

2 Setup

Data and forget sets. On Fashion-MNIST, 5,000 training images are held out for validation; D has 55,000 images. A forget set has $n = 600$ examples: *concentrated* (600 from one class c, i.e. 10.9% of that class; the other examples of c stay in D_r) or *stratified* (60 per class). $D_r = D \setminus D_f$, and conditions c0–c9 denote the ten concentrated sets. On CIFAR-10, $n = 490$ (49 per class when stratified).

Models and unlearning. Fashion-MNIST uses a CNN with three convolution–BatchNorm blocks and two linear layers, CIFAR-10 a CIFAR-style ResNet-18 with 20 BatchNorm layers; the original model O is trained with SGD and a cosine schedule for 15 (CNN) or 30 (ResNet) epochs. NegGrad+ (NG) minimizes $\alpha\,\mathrm{CE}_r - (1-\alpha)\,\mathrm{CE}_f$ for one epoch over D_r (425 steps on Fashion-MNIST) with SGD, learning rate 0.01, momentum 0.9, weight decay $5 \cdot 10^{-4}$ and global gradient clipping at 5, with $1-\alpha \in \{0.001, 0.01, 0.05\}$; the main weight is 0.01. Random-label descent (RL) replaces $-\mathrm{CE}_f$ by the cross-entropy to labels drawn once per seed among the other classes, balanced within each true class; it is a comparative control, not an equivalent unlearning method. Fine-tuning on D_r (FT) and a model retrained on D_r (R) serve as references.

BatchNorm treatment. In implementation **A**, retain and forget batches go through separate training-mode forward passes, so a concentrated forget batch is normalized by its own single-class statistics. Variant **C** puts all BatchNorm layers in evaluation mode with O's buffers; **E1** normalizes the forget batch with running buffers updated by retain batches only; **E1D** uses the forget batch's own statistics, as A does, but detaches them and does not update the buffers; **B** concatenates retain and forget batches. None of these comparisons changes only the source of the statistics: B also changes the retain statistics and couples the gradients, E1 also removes the gradient through the statistics, and E1D changes only the gradient path.

Measures. *Divergence*: maximum per-step $\mathrm{CE}_f > 10$ or a numerical failure, so a threshold crossing need not be a numerical failure; it is applied to NG only, since under RL the true-label loss rises by design. *Collapse*: test accuracy $\leq 20\%$ or one class receiving $\geq 80\%$ of test predictions. Collapse is measured with the BatchNorm buffers left by unlearning (native buffers). Under A, the forget batches contaminate these buffers: after exact recalibration on D_r, almost none of the collapsed runs stayed at $\leq 20\%$ (0 of 75 NG-A collapses in the RD64 study, 1 of 65 NG-def collapses in the tuning study, 4 of 70 NG collapses at 0.01 in the CF study; mean recalibrated test accuracy 70–74%). Collapse thus mostly marks a model unusable as deployed because of corrupted buffers; the damage that remains in the weights is measured by U. *Retained-utility loss U*: accuracy of O on D_r minus that of the unlearned model, both with BatchNorm buffers recalibrated on the same D_r (CF, RD64 and tuning studies; earlier studies count utility as kept when native-buffer test accuracy is within 5 points of O). *Force E*: mean drop of $\log_{10} p(y \mid x)$ on D_f relative to O, with recalibrated buffers; it measures a change in predictions, not forgetting. $\bar{G}$: mean over steps of $\log_{10}(\|g_t\|_2 + 10^{-8})$, where $\|g_t\|_2$ is the global norm of the combined gradient before clipping; it is the log of a geometric mean, not an instability measure. A *concentration penalty* Δ is, per seed, the mean of a measure over c0–c9 minus its value on the stratified set. The seed is the independent unit; confidence intervals are Student intervals over seeds, and Holm correction is applied within each prespecified family.

3 Concentration and NegGrad+ instability

Initial failure. The original study met its prespecified directional criterion only through collapsed runs: 26 of 55 runs ended at $\leq 20\%$ test accuracy with native buffers, all with concentrated forget sets (no recalibrated evaluation exists for that study). We report it as an instability of the method under that grid, not as evidence of forgetting.

Replication (protocol-specified, exploratory). A replication with newly derived seeds, draws and models compared A and C on all ten classes and three weights. Its statistic U_{rep} is the share of concentrated runs more unstable than the stratified run of the same seed, implementation and weight (a numerical failure counts as more unstable; otherwise the larger mean $\log_{10} \mathrm{CE}_f$); it is not a Mann–Whitney statistic. On Fashion-MNIST, U_{rep} was 0.86 under A ($p_{\mathrm{Holm}} = 1.7 \cdot 10^{-5}$) and the A–C difference was 0.37 ($p_{\mathrm{Holm}} = 1.5 \cdot 10^{-4}$). On CIFAR-10, U_{rep} was 1 under A in all three seeds, so its test was not computable, and the A–C difference was 0.43 ($p_{\mathrm{Holm}} = 0.0039$). Table 2 gives the counts: divergence and collapse are distinct, since on CIFAR-10 A diverged in 60 runs and never collapsed. With 5 and 3 seeds the protocol classifies these results as exploratory, and the analysis code was not committed before execution. At 0.01 on Fashion-MNIST, the six classes that diverged under A were those that collapsed in v3, so class identity and concentration were confounded.

Concentration beyond class identity (prospective). To separate the two, each of ten new seeds (with its own O) ran 68 compositions at $n = 600$ and $1 - \alpha = 0.01$: the 10 single-class sets, all 45 two-class sets, 12 five-class sets and the stratified set. Concentration is the label entropy $H_g \in \{0, 1, 2.32, 3.32\}$ bits. The outcome is the early forget loss, the mean of $\log_{10} \mathrm{CE}_f$ over steps 0–23, chosen beforehand on earlier traces where it separated runs that later diverged (AUC 0.92); it measures the early loss level, not

Study	Status	Seeds	Sec.
Original study v3	criterion prespecified	5	3
Diagnostics 2–3	post hoc	5	4, B
Replication v1.3	protocol-spec., exploratory	5; 3	3
Concentration (68 compositions)	prospective, exploratory	10 new	3
NG vs RL, clipping burden	prospective	10 (5 reused)	3
NG vs RL, $\bar{G}$	prospective	10 new	3
Frozen target (CF)	protocol-specified	12 new	4, D
Grouped RD64	protocol-specified	14 new	4
Per-condition tuning	protocol-specified	3 + 12 new	4
BatchNorm study	protocol-spec., exploratory	5	4, B
Statistics source (v1.2)	protocol-spec., exploratory	10 new	C
Control of C (CIFAR-10)	protocol-spec., exploratory	3	B
Tiny ImageNet, $\bar{G}$	prospective, exploratory	5 new	F

Table 1: Studies and their status. "Exploratory" follows each protocol's own classification given its seeds. No study is a public preregistration: dates rest on files or commits in a private repository. Shared data, splits and some seeds mean that later studies are not independent replications. Full provenance: Appendix A.

	Concentrated: diverged / collapsed		
Data, impl.	$1 - \alpha = 0.001$	0.01	0.05
FMNIST, A	5 / 0 (of 50)	30 / 24	50 / 47
FMNIST, C	0 / 0	0 / 0	18 / 0
CIFAR-10, A	0 / 0 (of 30)	30 / 0	30 / 0
CIFAR-10, C	0 / 0	0 / 0	2 / 0

Table 2: Replication v1.3 (protocol-specified, exploratory). Denominators: 50 concentrated runs per cell on Fashion-MNIST (10 classes × 5 seeds), 30 on CIFAR-10. Stratified runs never diverged or collapsed.

Classes in D_f	1	2	5	10
A, diverged	62/100	67/450	0/120	0/10
A, collapsed	54/100	22/450	0/120	0/10
A, utility kept	37/100	383/450	120/120	10/10
C, diverged / collapsed	0/100	0/450	0/120	0/10
C, utility kept	100/100	450/450	120/120	10/10

Table 3: Concentration study (seeds 401–410, $1-\alpha = 0.01$): descriptive counts. Utility kept: native-buffer test accuracy within 5 points of O. No run failed numerically.

divergence. Per seed we fitted

$$S(g) = \log_{10} M_\lambda\big(p_g;\, 10^\beta\big) + \gamma H_g + \varepsilon,$$

where M_λ is the power mean of order $\lambda \in [-3, 3]$ of the ten class effects β, weighted by the class proportions p_g; $\lambda \to 0$ and $\lambda = 1$ are additive combinations on the log and loss scales. The prediction was $\gamma < 0$, tested over seeds with a bootstrap-calibrated threshold. Under A, $\gamma = -0.246$ decade per bit (95% CI $[-0.263, -0.228]$, $p = 0.001$; all ten seeds negative; estimated λ 0.34–1.09): less concentrated sets had a lower early loss than their class composition predicts, for every combination rule in M_λ. Descriptively (Table 3), A diverged in 62/100 single-class and 67/450 two-class runs and never with five or ten classes, while C never diverged or collapsed. Under C (secondary), $\gamma = -0.004$ $[-0.016, 0.008]$, equivalent to zero at ± 0.075 at the estimated λ but positive with λ fixed at 0. The result does not exclude pair-specific class interactions, and it concerns one weight, one model and the first 24 steps.

Stronger under NegGrad+ than under random labels (prospective). The two objectives shared O, forget sets, batch streams, forward passes, BatchNorm treatment A and optimizer settings; per seed, $\theta = \Delta_{\mathrm{NG}} - \Delta_{\mathrm{RL}}$. On the share of clipped steps, $\theta_B = 0.304$ (95% CI $[0.289, 0.318]$) at 0.01, but RL sat at its floor (4 clipped steps in 330 runs), so this measure could not resolve smaller shifts. A second study on ten new seeds therefore used $\bar{G}$, with its analysis calibrated by Monte Carlo before any run. At 0.01, $\theta_{\bar{G}} = 0.652$ decade (95% CI $[0.607, 0.696]$, $p < 10^{-10}$), positive in all ten seeds (Table 4). RL showed a small shift (0.053) that its clipping burden had hidden, which does not imply RL instability. The direction was not the same at every weight, so $\bar{G}$ is not an instability measure. Concentrated NG runs lost 1.3, 46.8 and 82.9 test-accuracy points relative to O at the three weights, against 1.0, 2.0 and 2.9 under RL.

Beyond Fashion-MNIST (exploratory). A prospective extension to Tiny ImageNet [7], derived from ImageNet [6], used the CIFAR-style ResNet-18, five new seeds and ten target classes fixed before the archive was downloaded, with concentrated sets of 50 examples and paired stratified sets, at $1-\alpha = 0.01$ only. The contrast was again positive, $\theta_{\bar{G}} = 0.0288$ decade (95% CI $[0.0258, 0.0319]$), in all five seeds and all ten classes, although much smaller; the divergence criterion was saturated in both groups, and the archive's usage conditions were not verified with its host (Appendix F). With five seeds and one weight, this is not a powered replication.

	$1-\alpha = 0.001$	0.01	0.05
Δ_{NG}	0.009 [−.001, .019]	0.705 [.663, .746]	2.557 [2.509, 2.605]
Δ_{RL}	0.023 [.016, .031]	0.053 [.045, .061]	−0.087 [−.104, −.070]
$\theta_{\bar{G}}$	−0.015 [−.020, −.008]	0.652 [.607, .696]	2.644 [2.583, 2.705]
NG div. / coll.	9 / 0	60 / 53	100 / 100

Table 4: $\bar{G}$ study, seeds 101–110: concentration shift of $\bar{G}$ (decades), mean and 95% CI. Primary: $\theta_{\bar{G}}$ at 0.01; the other contrasts are Holm-corrected secondaries or descriptive. Bottom row: concentrated NG runs out of 100; RL never collapsed; stratified runs never diverged or collapsed.

Study (seeds)	Arm	Penalty Δ_U	Div.	Collapse
Frozen target (421–432)	NG	16.49	n.r.	70/120
	RL	0.23	—	0/120
	CF	0.25	—	0/120
Grouped RD64 (433–446)	NG-A	13.8	84/140	75/140
	NG-RD	0.2	0/140	0/140
Tuning (450–461)	NG-def	14.2	71/120	65/120
	NG-sel	1.07	2/120	0/120

Table 5: Concentration penalty on retained utility (points) and concentrated divergence and collapse counts at $1-\alpha = 0.01$ (NG-sel: per-condition configuration). Each block is a separate study. Collapse uses native buffers (Section 2). Divergence is not applied to RL or CF; n.r.: not reported in this summary. No run failed numerically.

4 Can the penalty be avoided?

A deployer facing a concentrated deletion request has three obvious levers: change how BatchNorm treats the forget batch, tune the optimizer, or change the forget objective. We report what each did, with Table 5 and Figure 1 summarizing the retained-utility penalty.

BatchNorm variants. If the instability came only from how the forget batch is normalized, changing that normalization should remove it at no cost; it did not do so consistently (Appendix B). In a post hoc diagnostic, B and C avoided all 29 of A's 30 divergences at 0.01 with utility within about 1.5 points of O. In a protocol-specified study with 150 concentrated and 50 stratified runs per weight, however, E1 diverged in 0, 60 and 150 concentrated runs at 0.001, 0.01 and 0.05, and in 47 stratified runs at 0.05; E1D diverged in 12, 89 and 121 concentrated runs and A in 12, 90 and 150. On the optimized forget loss, A exceeded E1D in a share of 0.82 of concentrated units ($p_{\mathrm{Holm}} = 3.7 \cdot 10^{-5}$), with a gap that grew with the weight, compatible with a weight-dependent contribution of the backward pass through the batch statistics that does not account for all of the instability. On CIFAR-10, C did

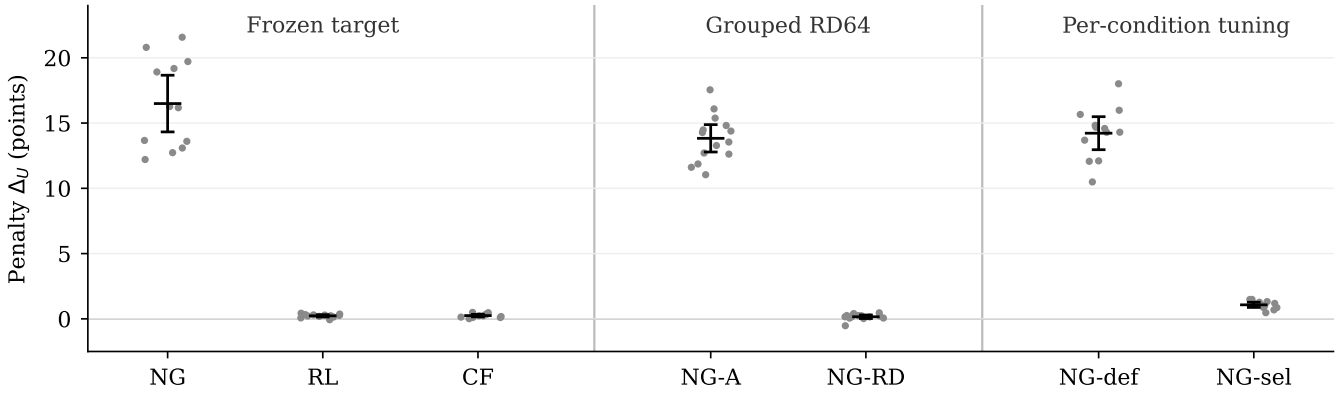


Figure 1: Concentration penalty on retained utility, Δ_U (points; concentrated mean over c0–c9 minus stratified), at $1-\alpha = 0.01$, in three separate pre-specified studies. Grey points: seeds; black: mean and 95% CI. Rows can be compared only within a study. A low penalty is not evidence of forgetting: RL, CF and NG-sel acted weakly on most concentrated sets (Section 4).

not diverge at 0.001 or 0.01 but cost 1.6–2.9 points relative to FT with native buffers, a cost removed by exact recalibration. These are associations between complete implementations and outcomes, not a decomposition of BatchNorm.

Grouped RD64 intervention. NG-RD normalizes the forget batch at each BatchNorm layer with the detached mean and biased variance of the first 64 rows of the concurrent retain batch, without updating buffers from the forget forward. It therefore changes three things together, the statistics source, the gradient path and the buffer updates, and no effect is attributed to the statistics alone (a separate early-loss study of the source alone is in Appendix C). A pre-specified study paired 14 new seeds; the number was fixed by a power simulation that allowed for a negative correlation between the two penalties (12 seeds gave 0.737, 14 gave 0.818–0.872 at five points). The single test on $I_{\mathrm{NG}} = \Delta_A - \Delta_{\mathrm{RD}}$ gave 13.7 points (95% CI $[12.6, 14.7]$, $p = 4 \cdot 10^{-13}$), positive in all 14 seeds (10.7–17.1). NG-RD neither diverged nor collapsed, against 84/140 and 75/140 for NG-A. Predictions on concentrated D_f also changed relative to FT ($F_{\mathrm{RD}} = 1.7$ points, $[1.5, 1.9]$): with recalibrated buffers, D_f accuracy was 80.2% for NG-A, 96.7% for NG-RD, 98.5% for FT and 93.1% for R. NG-RD's force was small ($E \approx 0.03$ decade); it is a change from FT, not evidence of forgetting.

The statistics source alone. A separate protocol-specified early-loss study (Appendix C) compared E1D and RD64, which differ only in the source of the detached statistics. Replacing the forget batch's own statistics by concurrent retain statistics raised the early loss burden of mixed compositions relative to single-class ones by 0.670 decade (95% CI $[0.632, 0.708]$), mostly by lowering it for single-class sets, and no RD64 run diverged (0/680, against 61/100 single-class E1D runs). A test meant to isolate concentration in this comparison failed its calibration and was withdrawn, so no concentration-specific effect or mechanism is claimed.

Per-condition tuning. A practitioner would tune hyperparameters, so a pre-specified study asked whether the penalty persists when the learning rate $\{0.001, 0.003, 0.01, 0.03\}$ and $1-\alpha$ $\{0.001, 0.003, 0.01, 0.03, 0.05\}$ are selected separately for each of the 11 conditions. On three tuning seeds, a configuration was *stable* if none of its runs failed, diverged or collapsed, and *feasible* if, in addition, its mean force E reached the mean force E_R of the retrained model. The rule chose the feasible configuration with the highest validation accuracy (buffers recalibrated on D_r), or else the strongest stable one, without seeing test data or evaluation seeds; the analysis recomputed the selection and found it identical. Twelve new evaluation seeds, fixed by power simulation, then ran the selected configuration (NG-sel) and the default $(0.01, 0.01)$ (NG-def).

The concentration penalty under NG-sel was $\Delta_{\mathrm{sel}} = 1.07$ points (95% CI $[0.88, 1.27]$, $p = 1.2 \cdot 10^{-7}$), equivalent to zero within the conventional ±2-point margin (TOST, 90% CI $[0.91, 1.24]$), against 14.2 $[13.0, 15.5]$ under NG-def; NG-sel diverged in 2/120 concentrated runs and never collapsed. Tuning was, however, *incomplete* (Table 6): the stratified set was stable under all 20 configurations, concentrated classes under 1 to 16, and only c4, c8 and c9 had a feasible configuration. For the other seven classes the rule fell back on the strongest stable configuration, which acted less than R, in six of them with $1-\alpha \leq 0.003$. Δ_{sel} is therefore not an equal-force comparison; on the evaluation seeds, $E \geq E_R$ held in 20% of concentrated and 67% of stratified runs.

A frozen alternative target. A third objective, CF, descends towards a target built once from O: for each forgotten (x, y), the softmax of O's logits over the classes other than y. In a pre-specified study with 12 new seeds, the interaction between concentration and objective, NG versus CF, was 16.2 points (95% CI $[14.1, 18.4]$, $p_{\mathrm{Holm}} = 7 \cdot 10^{-9}$); RL and CF did not differ detectably (-0.02, $[-0.07, 0.03]$, without an equivalence test), and CF's own penalty, 0.25 $[0.15, 0.35]$,

Condition	lr	$1-\alpha$	Stable	E / E_R	Status
stratified	0.01	0.03	20/20	0.092 / 0.085	feasible
c0	0.001	0.03	11/20	0.107 / 0.133	infeasible
c1	0.03	0.001	6/20	0.006 / 0.016	infeasible
c2	0.03	0.001	8/20	0.134 / 0.140	infeasible
c3	0.03	0.001	12/20	0.051 / 0.100	infeasible
c4	0.03	0.001	7/20	0.144 / 0.136	feasible
c5	0.01	0.003	9/20	0.004 / 0.027	infeasible
c6	0.03	0.003	10/20	0.135 / 0.242	infeasible
c7	0.001	0.001	1/20	0.002 / 0.028	infeasible
c8	0.001	0.05	14/20	0.144 / 0.017	feasible
c9	0.003	0.05	16/20	0.046 / 0.036	feasible

Table 6: Per-condition tuning: selected configuration, number of stable configurations out of 20, and mean force of the selected configuration versus the retrained model's force (decades), over the three tuning seeds.

was detected but within ±2 points. NG collapsed in 70/120 concentrated runs at 0.01 and 117/120 at 0.05, CF and RL never. The prespecified force check, however, rested on a nearly null reference (NG force 0.019 decade, CF 0.051), and on concentrated sets CF left D_f accuracy at 96.7–97.4% (FT 98.2%) with a force of 0.02–0.03 decade, although it fitted its targets during training. The utility advantage is therefore not successful unlearning (Appendix D).

5 Lessons for deployment

Within the configurations tested, four precautions follow; they are not guarantees.

1. **Inspect the class composition of deletion requests.** A request concentrated in one class was the regime in which NegGrad+ failed; stratified requests of the same size did not.
2. **Test an unlearning method on concentrated sets before deploying it.** Stratified forget sets of the same size never showed the failure, so an evaluation on them alone would have missed it.
3. **Report divergence, collapse, retained utility and force separately.** The original study's success criterion was met through collapsed runs, and collapse itself mostly reflected corrupted BatchNorm buffers; conversely, a stable run may change little, and CF and tuning preserved utility partly by acting weakly.
4. **Do not assume that hyperparameter tuning suffices.** Per-condition tuning removed most of the utility penalty, but for most concentrated classes only with configurations that did not reach the retrained model's force without instability.

6 Limitations and conclusion

What the evidence does not support. Neither intervention identifies a mechanism. RD64 changes the statistics source, the gradient path and the buffer updates together, and its utility gain came with a small force on concentrated sets ($E \approx 0.03$ decade). Δ_{sel} is not an equal-force comparison, and CF's force check used a near-zero reference. The concentration effect is defined relative to the M_λ family and does not exclude pair-specific class interactions. A low utility penalty, a stable run, proximity to R or a membership-inference AUC below 0.5 are not evidence of forgetting, erasure or privacy.

Limitations. Evidence is dominated by one small CNN and Fashion-MNIST; CIFAR-10 changes data and architecture together, and Tiny ImageNet is exploratory. RD64 and tuning each use one model, one dataset, one forget-set size and one epoch; tuning uses one 20-point grid and a force target fixed by R. Seed counts range from 3 to 14; data, splits and some seeds recur, and later hypotheses were informed by earlier results, so the studies are not independent replications. The concentration result concerns one weight and the first 24 steps; pair-specific interactions remain possible. A single retrained model was used per condition, so retraining variability is unknown. E, F_{RD}, proximity to R and membership-inference AUCs measure behavior change, not erasure, and no study assessed privacy.

Conclusion. Concentrating a fixed-size forget set in one class was associated with NegGrad+ instability in BatchNorm networks, with an early-loss effect beyond class composition and a gradient-norm shift much larger than under random-label descent. A bundled BatchNorm intervention and per-condition tuning reduced the retained-utility penalty, but tuning could not reach the retrained model's force without instability for most concentrated classes. Forget-set composition should therefore be treated as a test condition for unlearning in deployment, not as a detail of the benchmark.

References

[1] K. Zhao, M. Kurmanji, G.-O. Bărbulescu, E. Triantafillou, P. Triantafillou. What Makes Unlearning Hard and What to Do About It. arXiv:2406.01257.

[2] C. Fan, J. Liu, A. Hero, S. Liu. Challenging Forgets: Unveiling the Worst-Case Forget Sets in Machine Unlearning. arXiv:2403.07362.

[3] A. Kalani, M. Mandal, D. Kumar, M. Kankanhalli, Y. Sinha. The BatchNorm Illusion: Diagnosing Normalization Artifacts in Machine Unlearning Evaluation. arXiv:2609.08901.

[4] R. Zhang, L. Lin, Y. Bai, S. Mei. Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning. arXiv:2404.05868.

[5] I. Mavrothalassitis, P. Puigdemont, N. I. Levi, V. Cevher. Ascent Fails to Forget. arXiv:2509.26427.

[6] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei. ImageNet: A Large-Scale Hierarchical Image Database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

[7] Y. Le, X. Yang. Tiny ImageNet Visual Recognition Challenge. Stanford CS231N course project report, 2015.

A Provenance of each study

Table 7 lists each study with its provenance; Section 2 gives the common setup.

RD64 study. The prudent power scenario used a correlation $\rho = -0.5$ between Δ_A and Δ_{RD} and required power ≥ 0.80 at five points under both simulated laws. Type-I calibration gave rejection rates 0.042–0.056. With recalibrated buffers, the absolute distances to R in concentrated D_f accuracy were 14.4, 3.7 and 5.4 points for NG-A, NG-RD and FT: descriptively, NG-RD stayed closer to R, which does not establish forgetting.

Tuning study. Tuning used seeds 447–449 and evaluation seeds 450–461. The sole estimand was the per-seed concentration penalty under NG-sel, tested by a two-sided t-test (11 df) with TOST at ±2 points; between-seed SD was 0.31. Calibration was conforming (rejection 0.041–0.048; false equivalence 0.043–0.047). Descriptively, tuning reduced the penalty by 13.2 points (95% CI $[12.0, 14.3]$). $E \geq E_R$ is a tuning criterion averaged over tuning seeds, not forgetting, erasure or privacy.

B BatchNorm interventions

Table 8 assembles, post hoc, what each variant did in the studies in which it was run. Each study keeps its own design, denominators and utility measure; the rows are not comparable across studies and the table is not a confirmatory analysis.

- **C** did not diverge at 0.001 and 0.01 in any study, and diverged at 0.05 in 18/50 (Fashion-MNIST) and 2/30 (CIFAR-10) concentrated runs. On Fashion-MNIST its test accuracy stayed within about one point of O at the two smaller weights. On CIFAR-10/ResNet-18 it cost 1.6–2.9 points relative to FT at those weights; in the control study this cost was not removed by buffers tracked during unlearning and used only at evaluation, but was removed by exact recalibration.
- **E1** avoided divergence on c1 at 0.01 in diagnostic 3. In the protocol-specified BatchNorm study, it did not diverge at 0.001 but diverged in 60/150 concentrated runs at 0.01 and 150/150 at 0.05, and in 47/50 stratified runs at 0.05. On ResNet-18 it diverged in 31/33 cells at 0.01.
- **B** avoided divergence in all 30 tasks of diagnostic 2, at the frozen weight only.

BatchNorm study (protocol-specified, 5 seeds, exploratory). Its outputs were re-imported from the project drive and its four tests recomputed from the per-unit table; all values below match. It ran 600 units (450 concentrated, 150 stratified; 3 draws per class) with no numerical failure. On the optimized forget loss, A was above E1D in a share of 0.82 of concentrated units on average over seeds ($p_{\mathrm{Holm}} = 3.7 \cdot 10^{-5}$; mean gap +0.83 decade, 95% CI $[0.81, 0.86]$), and this share exceeded that under stratified sets by 0.33 ($p_{\mathrm{Holm}} = 4.0 \cdot 10^{-4}$). The gap grew with the weight (+0.01, +0.75 and +1.73 decade at 0.001, 0.01 and 0.05). For E1D versus E1, the ordinal test was significant but the continuous gap was +0.04 $[-0.08, 0.16]$, within the ±1-decade equivalence margin; the interaction for this contrast came mainly from E1's instability under stratified sets at 0.05. These results are compatible with a weight-dependent contribution of the standard backward pass through the batch statistics, which does not account for all of the instability, since E1D and E1 still diverged widely; the forward statistics showed no consistent average effect. The 60/150 E1 events at 0.01 are threshold-based divergences, not numerical failures. The study shares its split, classes and weights with v1.3, so it is a prospective follow-up rather than an independent replication.

How to read these results. They are associations between a complete implementation and an outcome, not a decomposition of the BatchNorm computation. C changes the source of the statistics and can also change the numerical trajectory through the retain and forget passes. The retain-only-buffer evaluation changes the measurement without changing the training trajectory, so its utility improvement shows evaluation sensitivity, not more forgetting.

C Source of the forget-batch statistics (v1.2)

E1D and RD64 differ only in the source of the detached statistics that normalize the forget batch in this early-loss study. On seeds 411–420, with the 68 compositions and the weight of the concentration study, the outcome is the capped early loss burden L_{20}, the mean over steps 0–19 of $\log_{10} \min(\max(\mathrm{CE}_f, 10^{-7}), 10)$. The primary estimand, fixed before the run, is

$$\theta_s = \tfrac{1}{3} \textstyle\sum_{k \in \{2,5,10\}} \big(\bar{D}_{k,s} - \bar{D}_{1,s}\big),$$

where $\bar{D}_{k,s}$ is the mean over the compositions with k classes of $L(\mathrm{RD64}) - L(\mathrm{E1D})$ for seed s (Student t, 9 df; H1 on the training forward pass, then H2 on a reference forward pass with retain-only buffers). Before the run, a simulation calibrated this test (false positives 0.045–0.051; power 0.95 at $\theta = 0.22$). It also showed that the coefficient $\Delta\gamma$ of the concentration model, meant to separate concentration from the combination rule, gave 8.2–19.3% false positives under combination forms outside M_λ; it was therefore withdrawn as a test and is reported only descriptively.

Results: $\theta = 0.670$ decade (95% CI $[0.632, 0.708]$, $p = 2.0 \cdot 10^{-11}$) on the training forward pass and 0.210 $[0.187, 0.232]$ ($p = 5.6 \cdot 10^{-9}$) on the reference pass, positive in all ten seeds and under both prespecified alternative weightings. In the protocol's wording: in this configuration, replacing the forget batch's own statistics by concurrent retain statistics raises the early loss burden of mixed compositions relative to pure ones, during training and at the ref-

Study	Status	Data, model	Seeds	Provenance
v3	criterion prespecified	FMNIST, CNN	5	Dated preregistration file; the directional criterion was met only through collapsed runs.
Diagnostics 2–3	**post hoc**	FMNIST, CNN	5	Classes chosen after v3; reading criteria dated before each computation.
Replication v1.3	**protocol-specified**	FMNIST CNN; CIFAR-10 ResNet-18	5; 3	Protocol committed before the run; analysis code *not* committed before execution.
BatchNorm study	**protocol-specified**, exploratory	FMNIST, CNN	5	Protocol and notebooks committed before the pilot; two engineering amendments and the inclusion decision committed before the main study. Shares the split, classes and weights of v1.3 and its hypotheses with diagnostic 3: not an independent replication.
Concentration study	**prospective**, exploratory	FMNIST, CNN	10 (new)	Protocol, notebook, analysis and tests committed before any run; post-pilot decision on timings only; frozen analysis run unchanged.
Statistics source	**protocol-specified**, exploratory	FMNIST, CNN	10 (new)	Protocol, notebook, analysis, tests and calibration committed and pushed before the pilot; one amendment (a unit-check tolerance) after a pilot stop that preceded any task; post-pilot decision on timings only; frozen analysis run unchanged.
Grouped RD64 intervention	**protocol-specified**	FMNIST, CNN	14 (new)	Protocol, notebook, analysis, tests, power and calibration committed and pushed before the pilot; post-pilot decision on timings only; frozen analysis run unchanged; no amendment.
Per-condition tuning	**protocol-specified**	FMNIST, CNN	3 + 12 (new)	Protocol, notebook, analysis, tests, power and calibration committed and pushed before the pilot; post-pilot decision on timings only; frozen analysis run unchanged; no amendment. Two Colab sessions (A100 40 GB then 80 GB) were recorded by the notebook and were non-blocking.
Frozen target (CF)	**protocol-specified**	FMNIST, CNN	12 (new)	Drafted with 10 seeds; moved to 12 before the freeze after the power simulation. Protocol, notebook, analysis, tests, power and calibration committed and pushed before the pilot; one amendment (a unit-test tolerance) after a pilot stop that preceded any task; post-pilot decision on timings only; frozen analysis run unchanged.
Control of C	**protocol-specified**, exploratory	CIFAR-10, ResNet-18	3	Plan, notebook and analysis frozen in one commit; pilot and study ran from copies built before the freeze (code identical except a version label); one committed amendment was not applied.
NG vs RL, clipping	**prospective**	FMNIST, CNN	10	Protocol and analysis committed before the pilot; notebook committed after the pilot (identical copy); seeds 1–5 reuse v1.3 models.
NG vs RL, $\bar{G}$	**prospective**	FMNIST, CNN	10 (new)	Protocol, analysis, tests and Monte Carlo calibration committed before any run; notebook committed separately before the pilot; one amendment (a unit-test tolerance) after a pilot stop that preceded any unlearning run.
Tiny ImageNet, $\bar{G}$	**prospective**, exploratory	Tiny ImageNet, ResNet-18	5 (new)	Protocol, analysis, tests and calibration committed before any training; cost pilot ran from an uncommitted copy identical to the later-committed notebook; launch decision computed in the pilot's session (access log: only the allowed files read) and committed before the study. Usage conditions of the archive *not verified* with its host.

Table 7: Studies used in this paper, with their exact status and provenance. "Exploratory" follows each protocol's own classification given the number of seeds.

erence evaluation. Descriptively, RD64 lowered L_{20} by 0.79 decade for single-class sets, 0.31 for pairs, 0.06 for five-class sets and not at all for the stratified set. No RD64 run diverged or collapsed (0/680), against 61/100 single-class and 53/450 two-class E1D runs that diverged. $\Delta\gamma$ was +0.235 $[0.220, 0.251]$, without a p-value. The study does not isolate an effect of concentration from a change in how class effects combine, and it does not separate the source of the statistics from the mismatch between the forget batch and the statistics used, so it does not show that homogeneous statistics are a causal mechanism.

D Frozen alternative target (CF)

For each forgotten example (x, y), $q(\cdot \mid x)$ is the softmax of O's logits (evaluation mode, O's buffers) over the classes $k \neq y$, the true class being masked before the softmax, at temperature 1. CF minimizes $\alpha\,\mathrm{CE}_r + (1-\alpha)\,\mathrm{CE}(q, p_\theta)$. NG, RL and CF share O, the forget sets, the batch schedule and BatchNorm treatment A. NG ran at $1-\alpha \in \{0.01, 0.05\}$, RL and CF at $\{0.01, 0.05, 0.2\}$, with FT, R and O recalibrated for each condition, on seeds 421–432. Per seed, Δ_m is U averaged over c0–c9 minus U on the stratified set, and $I_{m,\mathrm{CF}} = \Delta_m - \Delta_{\mathrm{CF}}$; the two primary tests, $I_{\mathrm{NG,CF}}$ and $I_{\mathrm{RL,CF}}$ at 0.01, are t-tests over seeds (11 df) with Holm correction. The draft

Study (status)	Variant	Cells	Divergence (concentrated)	Utility
Diagnostic 2 (post hoc)	A / B / C	30 tasks, frozen weight (0.01)	A 29/30; B 0/30; C 0/30	B and C within ≈1.5 points of O
Diagnostic 3 (post hoc)	E1	class c1, 5 seeds, 0.01	0/5	92.4–93.3% (O 93.0–93.4%)
Replication v1.3 (protocol-spec.)	C	FMNIST, 50 per weight	0 / 0 / 18 at 0.001 / 0.01 / 0.05	mean test 92.4 / 92.4 / 90.1% (O 93.0–93.4%)
	C	CIFAR-10, 30 per weight	0 / 0 / 2	mean test 88.5 / 88.8 / 85.5% (O 92.4–92.7%)
BatchNorm study (protocol-spec.)	E1	FMNIST, 150 conc. / 50 strat. per weight	conc. 0 / 60 / 150; strat. 0 / 0 / 47	degraded (< O − 5 points), conc.: 0 / 60 / 150
	E1D	same cells	conc. 12 / 89 / 121; strat. 0 / 0 / 0	degraded, conc.: 0 / 104 / 150
	A	same cells	conc. 12 / 90 / 150; strat. 0 / 0 / 0	degraded, conc.: 10 / 90 / 150 (A's own buffers); 0 / 89 / 150 (retain-only buffers)
Control of C (protocol-spec.)	C	CIFAR-10, 33 cells at 0.001 and at 0.01	0/33 and 0/33	1.6–2.9 points below FT (native); not below FT after recalibration
	E1	same cells	0/33 and 31/33	at 0.001, at the FT level

Table 8: Post hoc, descriptive assembly of intervention results, separated by study. Each row is limited to the cells in which the variant was run; B was run only in diagnostic 2. Utility measures differ between studies and are given as defined in each. "Cells" on CIFAR-10 are condition × weight combinations over 3 seeds.

	Δ_U, 0.01	Δ_U, 0.05	Δ_U, 0.2	Collapses
NG	16.49 [14.32, 18.66]	47.96 [46.32, 49.59]	–	70 / 117 / –
RL	0.23 [0.15, 0.32]	−0.39 [−0.54, −0.23]	−1.77 [−2.03, −1.50]	0 / 0 / 0
CF	0.25 [0.15, 0.35]	−0.42 [−0.59, −0.25]	−1.46 [−1.62, −1.29]	0 / 0 / 0

Table 9: Frozen-target study, seeds 421–432. Δ_U: concentrated-minus-stratified utility loss on retained data (points), mean and 95% CI over 12 seeds; only Δ_{CF} at 0.01 is a prespecified (secondary) test. Collapses: concentrated runs at 0.01 / 0.05 / 0.2, out of 120 per cell. No run failed numerically.

planned 10 seeds; the power simulation gave 0.72, below the prespecified 0.80, so the protocol was set to 12 seeds (0.81) before the freeze. Family-wise type I rates had upper 95% bounds ≤ 0.068. All 1,467 units ran, and replays were bitwise identical.

- $I_{NG,CF}(0.01)$ = 16.2 points, 95% CI $[14.1, 18.4]$, $p_{Holm} = 7 \cdot 10^{-9}$, positive in all 12 seeds (12.1–21.2).
- $I_{RL,CF}(0.01)$ = −0.02 point $[-0.07, 0.03]$, p_{Holm} = 0.41: no difference detected. No equivalence test was prespecified for this contrast, so this is not evidence that the two objectives are equivalent.
- $\Delta_{CF}(0.01)$ = +0.25 point $[0.15, 0.35]$, detected ($p_{Holm} = 3 \cdot 10^{-4}$) and within the prespecified, exploratory equivalence margin of ±2 points (90% CI $[0.17, 0.33]$), a convention fixed before the run, not a threshold of practical relevance.

Force check. The prespecified rule selects the smallest CF weight whose force on the stratified set is at least that of NG at 0.01. It selected 0.01, but that reference force was nearly zero (E = 0.019 decade for NG; 0.051 for CF), so the check does not establish a comparison at a sufficient forgetting force. The prespecified test and its label are unchanged.

Post hoc observations. On concentrated sets, CF's accuracy on D_f (recalibrated buffers) stayed at 96.7–97.4% at all three weights, against 98.2% for FT, and its force was E = 0.02–0.03 decade, although CF fitted its targets in the training forward pass (final cross-entropy to q of 0.69–0.97). On the stratified set, CF at 0.2 lowered the accuracy on D_f to 10.5%; RL showed the same pattern (93.7–98.1% on concentrated sets, 25.8% on the stratified set at 0.2). This gap between fitting the training objective and the evaluated predictions is neither evidence of erasure nor a demonstration of a BatchNorm mechanism.

E Measurement details and descriptive counts

Concentration study. Each of the 12 five-class sets contains each class 6 times. The early loss is truncated before a divergence that would occur earlier (4 of 680 A runs, all single-class). The t threshold was calibrated by parametric bootstrap under $\gamma = 0$ (B = 2,000); the estimated λ had median 0.91.

NG versus RL. The $\bar{G}$ study (seeds 101–110) had type I error 0.044–0.059 under three null models, Holm family-wise rate 0.036–0.044 and false equivalence 0.047–0.052; all 782 units completed with bitwise-identical replays. Secondary tests (Holm, m = 3): Δ_{RL} at 0.01 and $\theta_{\bar{G}}$ at 0.001 and 0.05 ($p_{Holm} = 2.0 \cdot 10^{-7}$, 0.0004, $< 10^{-10}$); equivalence of Δ_{RL} to zero at ±0.02 decade was not established. The clipping study reused v1.3 models for seeds 1–5.

F Exploratory extension to Tiny ImageNet

Tiny ImageNet [7], derived from ImageNet [6], was used with the CIFAR-style ResNet-18 and a 200-class output, on five new seeds. For each of ten target classes fixed before the archive was downloaded, a concentrated set of 50 examples of the class is paired with a stratified set holding one example of the class and one of each of 49 other classes. The forget batch has 25 examples, so each forgotten example is presented 351–352 times in 703 steps; only $1-\alpha = 0.01$ was run. The primary contrast was $\theta_{\bar{G}} = 0.0288$ decade (95% CI $[0.0258, 0.0319]$, 4 df, $p = 1.3 \cdot 10^{-5}$), positive in all five seeds and all ten classes; $\Delta_{\mathrm{NG}} = 0.0232$ $[0.0212, 0.0252]$ and $\Delta_{\mathrm{RL}} = -0.0056$ $[-0.0069, -0.0043]$, detected ($p_{\mathrm{Holm}} = 3 \cdot 10^{-4}$) and within the ± 0.02 equivalence margin. No run collapsed or failed numerically; the NG divergence criterion was saturated (50/50 concentrated and 50/50 stratified runs). An exploratory membership-inference analysis on concentrated sets, with the fixed score $-$CE and official validation images as non-members, gave mean AUCs of 0.732 for O, 0.670 for FT, 0.590 for RL and 0.388 for NG; the NG value below 0.5 is an inversion of the score order, not evidence of erasure. The archive's usage conditions were not verified with its host; the study proceeded on the authors' decision, under the probable framework of the ImageNet access agreement (non-commercial research).